Multi-Source Emotion Classification for Analysis of Customer Feedback

Arihant Gupta
Enrolment No: E23CSEU0055

Sutejas Hashia
Enrolment No: E23CSEU0047

Samrat Singh
Enrolment No: E23CSEU0033

September 3, 2025

Abstract

This project aims to develop a sophisticated emotion classification system to analyze customer feedback from social media. Moving beyond simple sentiment polarity (positive, negative), our objective is to classify text into a richer set of discrete emotions such as Joy, Anger, and Sadness. To build a robust model, we will aggregate multiple public datasets, including domain-specific airline tweets and general emotion corpora. A key challenge is the creation of a unified emotion taxonomy to harmonize varied labels from different sources. The methodology involves advanced data preprocessing, TF-IDF feature extraction, and a comparative study of machine learning models, focusing on handling class imbalance. The resulting system will provide nuanced insights into the specific emotional responses of textual data.

Proposed Method

Our methodology is designed to handle the complexity of multi-source data aggregation and multi-class emotion classification.

The process begins with combining multiple datasets into a single, standardized format. The core of this step is to create a unified set of target emotion categories (e.g., Joy, Sadness, Anger, Fear, Surprise).

For Feature Extraction, we will convert the cleaned text into a numerical representation using the **TF-IDF (Term Frequency-Inverse Document Frequency)** technique. To capture crucial context, our TF-IDF model will include both unigrams (single words) and bigrams (two-word phrases). The formula is given by: Finally, for Model Implementation, we will establish a baseline using traditional classifiers like **Logistic Regression** and **Support Vector Machines (SVM)**. To capture deeper linguistic nuances essential for emotion detection, we will also explore a **Deep Learning approach** using a Long Short-Term Memory (LSTM) network. A key consideration is handling the inherent **class imbalance** in emotion data, which we will address using class weighting during model training. Model performance will be evaluated using Accuracy, per-class Precision, Recall, F1-Score, and a Confusion Matrix to analyze classification errors between similar emotions.

Datasets Used

The primary dataset is the Airline Tweet Sentiment collection, which contains over 14,000 airline-related tweets labeled as Positive, Negative, or Neutral. This provides the crucial domain-specific language and context for our primary use case.

To expand the model's emotional range, this dataset will be augmented with several publicly available emotion corpora. These include the GoEmotions dataset from Google, which contains Reddit comments labeled with 27 fine-grained emotions; the classic ISEAR (International Survey on Emotion Antecedents and Reactions) dataset; and the DailyDialog dataset, which provides emotion labels in a conversational context. By combining these varied sources, we will create a rich and diverse dataset for training a comprehensive and effective emotion classifier.